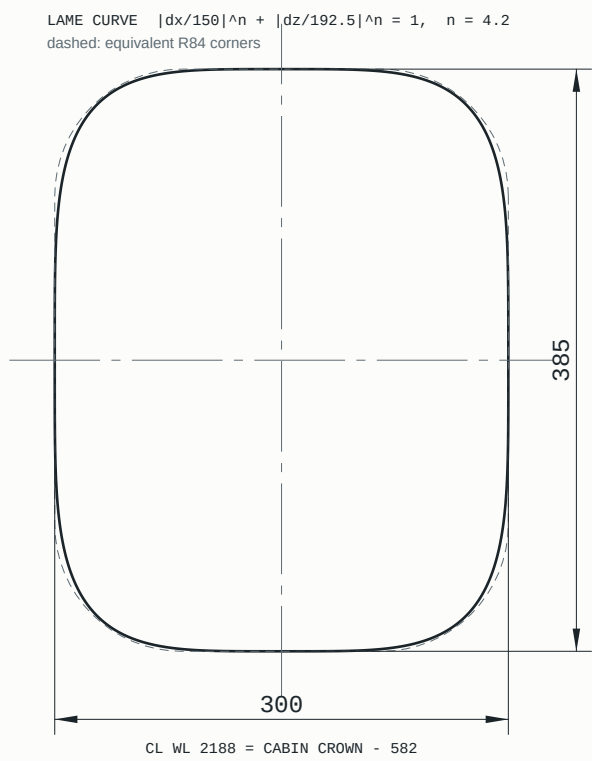
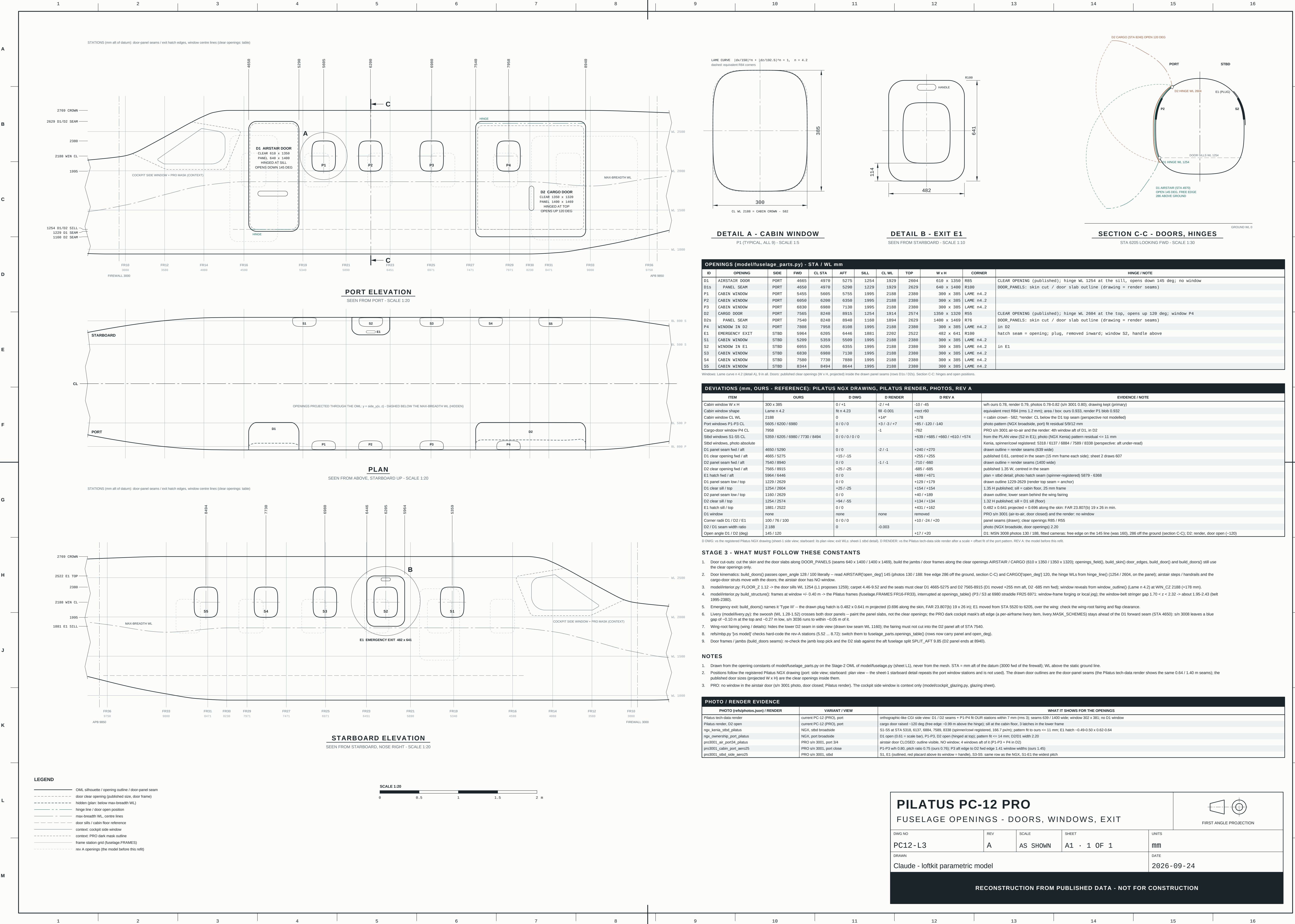
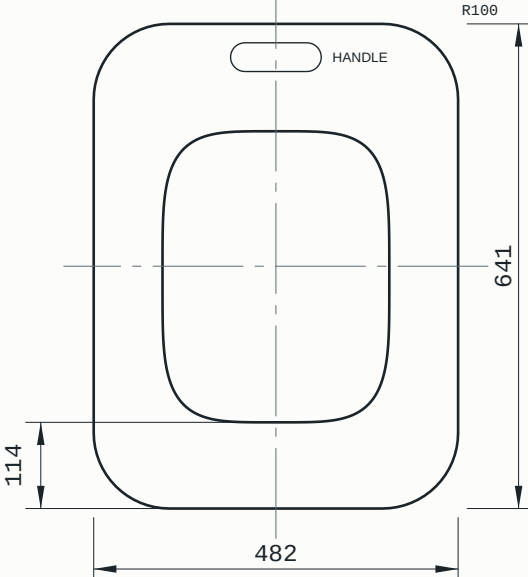




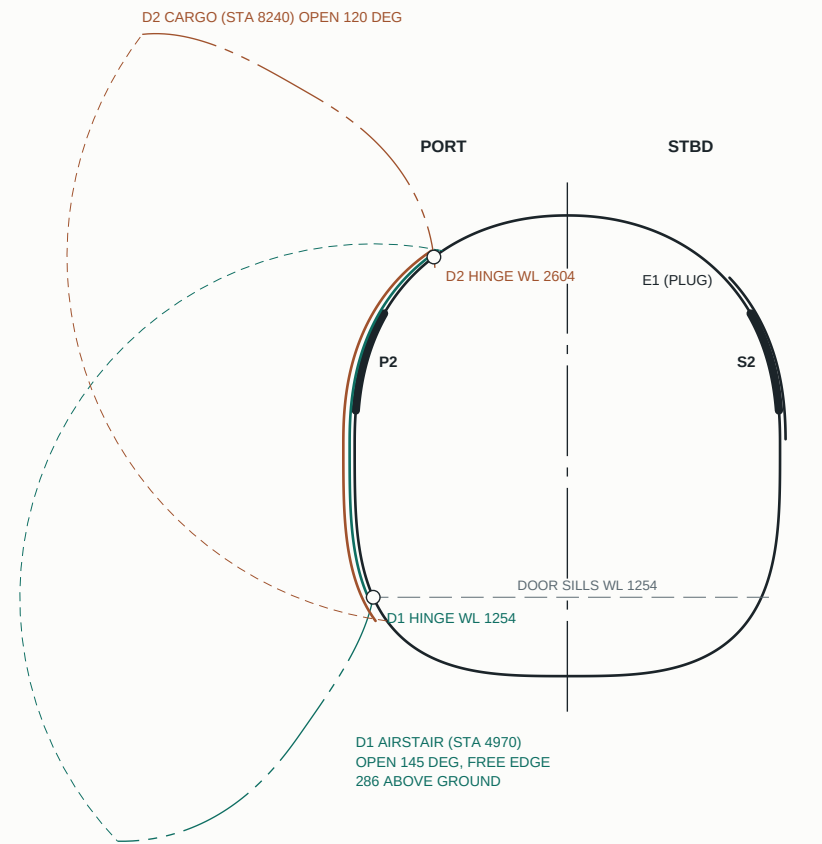
DETAIL A - CABIN WINDOW

P1 (TYPICAL, ALL 9) - SCALE 1:5



DETAIL B - EXIT E1

SEEN FROM STARBOARD - SCALE 1:10



SECTION C-C - DOORS, HINGES

STA 6205 LOOKING FWD - SCALE 1:30

| OPENINGS (model/fuselage_parts.py) - STA / WL mm |                |      |      |        |      |      |       |      |             |           |                                                                                     |
|--------------------------------------------------|----------------|------|------|--------|------|------|-------|------|-------------|-----------|-------------------------------------------------------------------------------------|
| ID                                               | OPENING        | SIDE | FWD  | CL STA | AFT  | SILL | CL WL | TOP  | W x H       | CORNER    | HINGE / NOTE                                                                        |
| D1                                               | AIRSTAIR DOOR  | PORT | 4665 | 4978   | 5275 | 1254 | 1929  | 2684 | 618 x 1358  | R85       | CLEAR OPENING (published); hinge WL 1254 at the sill, opens down 145 deg; no window |
| D1S                                              | PANEL SEAM     | PORT | 4658 | 4978   | 5298 | 1229 | 1929  | 2629 | 648 x 1408  | R188      | DOOR_PANELS: skin cut / door slab outline (drawing = render seams)                  |
| P1                                               | CABIN WINDOW   | PORT | 5455 | 5685   | 5755 | 1995 | 2188  | 2388 | 388 x 385   | LAME n4.2 |                                                                                     |
| P2                                               | CABIN WINDOW   | PORT | 6858 | 6288   | 6358 | 1995 | 2188  | 2388 | 388 x 385   | LAME n4.2 |                                                                                     |
| P3                                               | CABIN WINDOW   | PORT | 6838 | 6988   | 7138 | 1995 | 2188  | 2388 | 388 x 385   | LAME n4.2 |                                                                                     |
| D2                                               | CARGO DOOR     | PORT | 7565 | 8248   | 8915 | 1254 | 1914  | 2574 | 1358 x 1328 | R55       | CLEAR OPENING (published); hinge WL 2684 at the top, opens up 120 deg; window P4    |
| D2S                                              | PANEL SEAM     | PORT | 7548 | 8248   | 8948 | 1168 | 1894  | 2629 | 1488 x 1468 | R76       | DOOR_PANELS: skin cut / door slab outline (drawing = render seams)                  |
| P4                                               | WINDOW IN D2   | PORT | 7888 | 7958   | 8188 | 1995 | 2188  | 2388 | 388 x 385   | LAME n4.2 | in D2                                                                               |
| E1                                               | EMERGENCY EXIT | STBD | 5964 | 6285   | 6446 | 1881 | 2282  | 2522 | 482 x 641   | R188      | hatch seam = opening; plug, removed inward; window S2, handle above                 |
| S1                                               | CABIN WINDOW   | STBD | 5288 | 5358   | 5588 | 1995 | 2188  | 2388 | 388 x 385   | LAME n4.2 |                                                                                     |
| S2                                               | WINDOW IN E1   | STBD | 6855 | 6285   | 6355 | 1995 | 2188  | 2388 | 388 x 385   | LAME n4.2 | in E1                                                                               |
| S3                                               | CABIN WINDOW   | STBD | 6838 | 6988   | 7138 | 1995 | 2188  | 2388 | 388 x 385   | LAME n4.2 |                                                                                     |
| S4                                               | CABIN WINDOW   | STBD | 7588 | 7738   | 7888 | 1995 | 2188  | 2388 | 388 x 385   | LAME n4.2 |                                                                                     |
| S5                                               | CABIN WINDOW   | STBD | 8344 | 8484   | 8644 | 1995 | 2188  | 2388 | 388 x 385   | LAME n4.2 |                                                                                     |

Windows: Lame curve n 4.2 (detail A), 9 in all. Doors: published clear openings (W x H, projected) inside the drawn panel seams (rows D1S / D2S). Section C-C: hinges and open positions.

| DEVIATIONS (mm, OURS - REFERENCE): PILATUS NGX DRAWING, PILATUS RENDER, PHOTOS, REV A |                                  |                   |              |                                  |  | EVIDENCE / NOTE                                                                                                                                    |
|---------------------------------------------------------------------------------------|----------------------------------|-------------------|--------------|----------------------------------|--|----------------------------------------------------------------------------------------------------------------------------------------------------|
| ITEM                                                                                  | OURS                             | D DWG             | D RENDER     | D REV A                          |  |                                                                                                                                                    |
| Cabin window W x H                                                                    | 300 x 385                        | 0 / +1            | -2 / +4      | -10 / +5                         |  | with ours 0.78, render 0.79, photos 0.78-0.82 (s/n 3001 0.80); drawing kept (primary)                                                              |
| Cabin window shape                                                                    | Lame n 4.2                       | fit n 4.23        | fill -0.001  | rect #60                         |  | equivalent rect R84 (rms 1.2 mm); area / box: ours 0.933, render P1 blob 0.932                                                                     |
| Cabin window CL WL                                                                    | 2188                             | 0                 | +14*         | +178                             |  | = cabin crown - 582; *render: CL below the D1 top seam (perspective not modelled)                                                                  |
| Port windows P1-P3 CL                                                                 | 5605 / 6200 / 6980               | 0 / 0 / 0         | +3 / -3 / +7 | +85 / -120 / -140                |  | photo pattern (NGX broadside, port) fit residual 5/9/12 mm                                                                                         |
| Cargo-door window P4 CL                                                               | 7958                             | 0                 | -1           | -762                             |  | PRO s/n 3001 air-to-air and the render: 4th window aft of D1, in D2                                                                                |
| Sbd windows S1-S5 CL                                                                  | 5359 / 6205 / 6980 / 7730 / 8494 | 0 / 0 / 0 / 0 / 0 | -1           | +639 / +685 / +660 / +610 / +574 |  | from the PLAN view (S2 in E1); photo (NGX Kenya) pattern residual <= 11 mm                                                                         |
| Sbd windows, photo absolute                                                           |                                  |                   |              |                                  |  | Kenia, spinner/cowl registered: 5318 / 6137 / 6884 / 7589 / 8338 (perspective: alt under-read)                                                     |
| D1 panel seam fwd / aft                                                               | 4650 / 5290                      | 0 / 0             | -2 / -1      | +240 / +270                      |  | drawn outline = render seams (639 wide)                                                                                                            |
| D1 clear opening fwd / aft                                                            | 4655 / 5275                      | +15 / -15         |              | +255 / +255                      |  | published 0.61, centred in the seam (15 mm frame each side), sheet 2 draws 807                                                                     |
| D2 panel seam fwd / aft                                                               | 7540 / 8940                      | 0 / 0             | -1 / -1      | -710 / -680                      |  | drawn outline = render seams (1400 wide)                                                                                                           |
| D2 clear opening fwd / aft                                                            | 7565 / 8915                      | +25 / -25         |              | -685 / -685                      |  | published 1.35 W, centred in the seam                                                                                                              |
| E1 hatch fwd / aft                                                                    | 5964 / 6446                      | 0 / 0             |              | +699 / +671                      |  | plan = sbd detail; photo hatch seam (spinner-registered) 5879 - 6368                                                                               |
| D1 panel seam low / top                                                               | 1229 / 2629                      | 0 / 0             |              | +129 / +179                      |  | drawn outline 1229-2629 (render top seam = anchor)                                                                                                 |
| D1 clear sill / top                                                                   | 1254 / 2604                      | +25 / -25         |              | +154 / +154                      |  | 1.35 H published; sill = cabin floor, 25 mm frame                                                                                                  |
| D2 panel seam low / top                                                               | 1160 / 2629                      | 0 / 0             |              | +40 / +189                       |  | drawn outline; lower seam behind the wing fairing                                                                                                  |
| D2 clear sill / top                                                                   | 1254 / 2574                      | +94 / -55         |              | +134 / +134                      |  | 1.32 H published; sill = D1 sill (floor)                                                                                                           |
| E1 hatch sill / top                                                                   | 1881 / 2522                      | 0 / 0             |              | +431 / +162                      |  | 0.482 x 0.641 projected = 0.696 along the skin: FAR 23.807(b) 19 x 26 in min.                                                                      |
| D1 window                                                                             | none                             | none              | none         | removed                          |  | PRO s/n 3001 (air-to-air, door closed) and the render: no window                                                                                   |
| Corner radii D1 / D2 / E1                                                             | 100 / 76 / 100                   | 0 / 0 / 0         |              | +10 / -24 / +20                  |  | panel seams (drawn); clear openings R85 / R55                                                                                                      |
| D2 / D1 seam width ratio                                                              | 2.188                            | 0                 | -0.003       |                                  |  | photo (NGX broadside, door openings) 2.20                                                                                                          |
| Open angle D1 / D2 (deg)                                                              | 145 / 120                        |                   |              | +17 / +20                        |  | D1: MSN 3008 photos 130 / 188, fitted cameras: free edge on the 145 line (was 160), 286 off the ground (section C-C); D2: render, door open (-120) |

D DWG: vs the registered Pilatus NGX drawing (sheet-1 side view; starboard; its plan view; exit WLS: sheet-1 sbd detail). D RENDER: vs the Pilatus tech-data side render after a scale + offset fit of the port pattern. REV A: the model before this refit.

### STAGE 3 - WHAT MUST FOLLOW THESE CONSTANTS

- Door cut-outs: cut the skin and the door slabs along DOOR\_PANELS (seams 640 x 1400 / 1400 x 1469), build the jambs / door frames along the clear openings AIRSTAIR / CARGO (610 x 1350 / 1350 x 1320); openings\_field(), build\_skin() door\_edges, build\_door() and build\_doors() still use the clear openings only.
- Door kinematics: build\_doors() passes open\_angle 128 / 100 literally -- read AIRSTAIR[open\_deg] 145 (photos 130 / 188: free edge 286 off the ground, section C-C) and CARGO[open\_deg] 120, the hinge WLS from hinge\_line() (1254 / 2604, on the panel); airstair steps / handrails and the cargo-door struts move with the doors; the airstair door has NO window.
- modelInterior.py: FLOOR\_Z 1.12 -> the door sills WL 1254 (L1 proposes 1259); carpet 4.46-9.52 and the seats must clear D1 4665-5275 and D2 7565-8915 (D1 moved +255 mm aft, D2 -685 mm fwd); window reveals from window\_outline() (Lame n 4.2) at WIN\_CZ 2188 (+178 mm).
- modelInterior.py build\_structures(): frames at window +/- 0.40 m -> the Pilatus frames (fuselage.FRAMES FR16-FR33), interrupted at openings\_table() (P3 / S3 at 6980 straddle FR25 6971: window-frame forging or local jog); the window-belt stringer gap 1.70 < z < 2.32 -> about 1.95-2.43 (belt 1995-2380).
- Emergency exit: build\_doors() names it 'Type III' -- the drawn plug hatch is 0.482 x 0.641 m projected (0.696 along the skin, FAR 23.807(b) 19 x 26 in); E1 moved from STA 5520 to 6205, over the wing: check the wing-root fairing and flap clearance.
- Livery (model/livery.py): the swoosh (WL 1.28-1.52) crosses both door panels -- paint the panel slabs, not the clear openings; the PRO dark cockpit mask's aft edge (a per-airframe livery item, livery.MASK\_SCHEMES) stays ahead of the D1 forward seam (STA 4650); s/n 3008 leaves a blue gap of -0.10 m at the top and -0.27 m low, s/n 3036 runs to within -0.05 m of it.
- Wing-root fairing (wing / details): hides the lower D2 seam in side view (drawn low seam WL 1160); the fairing must not cut into the D2 panel aft of STA 7540.
- ref/smbp.py [vs model]: checks hard-code the rev-A stations (5.52 ... 8.72): switch them to fuselage\_parts.openings\_table() (rows now carry panel and open, deg).
- Door frames / jambs (build\_doors seams): re-check the jamb loop pick and the D2 slab against the aft fuselage split SPLIT\_AFT 9.85 (D2 panel ends at 8940).

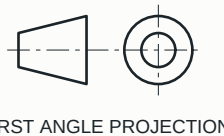
### NOTES

- Drawn from the opening constants of model/fuselage\_parts.py on the Stage-2 OML of model/fuselage.py (sheet L1), never from the mesh. STA = mm aft of the datum (3000 fwd of the firewall); WL above the static ground line.
- Positions follow the registered Pilatus NGX drawing (port: side view; starboard: plan view -- the sheet-1 starboard detail repeats the port window stations and is not used). The drawn door outlines are the door-panel seams (the Pilatus tech-data render shows the same 0.64 / 1.40 m seams); the published door sizes (projected W x H) are the clear openings inside them.
- PRO: no window in the airstair door (s/n 3001 photo, door closed; Pilatus render). The cockpit side window is context only (model/cockpit\_glazing.py, glazing sheet).

| PHOTO / RENDER EVIDENCE          |                           |                                                                                                                                                    |  |
|----------------------------------|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|--|
| PHOTO (ref/photos.json) / RENDER | VARIANT / VIEW            | WHAT IT SHOWS FOR THE OPENINGS                                                                                                                     |  |
| Pilatus tech-data render         | current PC-12 (PRO), port | orthographic-like CGI side view. D1 / D2 seams + P1-P4 fit OUR stations within 7 mm (rms 3); seams 639 / 1400 wide; window 302 x 381; no D1 window |  |
| Pilatus render, D2 open          | current PC-12 (PRO), port | cargo door raised -120 deg (free edge -0.99 m above the hinge); sill at the cabin floor, 3 inches in the lower frame                               |  |
| ngx_kenya_sbd_pilatus            | NGX, sbd broadside        | S1-S5 at STA 5318, 6137, 6884, 7589, 8338 (spinner/cowl registered, 166.7 pphm); pattern fit to ours <= 11 mm; E1 hatch -0.49-0.50 x 0.62-0.64     |  |
| ngx_ownership_port_pilatus       | NGX, port broadside       | D1 open (0.61 = scale bar), P1-P3, D2 open (hinged at top); pattern fit <= 14 mm; D2/D1 width 2.20                                                 |  |
| pro3001_ar_port34_pilatus        | PRO s/n 3001, port 3/4    | airstair door CLOSED: outline visible. No window, 4 windows aft of it (P1-P3 + P4 in D2)                                                           |  |
| pro3001_cabin_port_aero25        | PRO s/n 3001, port close  | P1-P3 with 0.80, pitch ratio 0.75 (ours 0.76); P3 aft edge to D2 fwd edge 1.41 window widths (ours 1.45)                                           |  |
| pro3001_sbd_side_aero25          | PRO s/n 3001, sbd         | S1, E1 (outlined, red placard above its window = handle); S3-S5: same row as the NGX. S1-E1 the widest pitch                                       |  |

## PILATUS PC-12 PRO

### FUSELAGE OPENINGS - DOORS, WINDOWS, EXIT



|            |                                   |          |             |       |
|------------|-----------------------------------|----------|-------------|-------|
| DWG NO     | REV                               | SCALE    | SHEET       | UNITS |
| PC12-L3    | A                                 | AS SHOWN | A1 · 1 OF 1 | mm    |
| DRAWN      | Claude - loftkit parametric model |          |             | DATE  |
| 2026-09-24 |                                   |          |             |       |

RECONSTRUCTION FROM PUBLISHED DATA - NOT FOR CONSTRUCTION